VoiceMate AI

One-Page Solution Pitch | The Great Indian AI Internship Challenge

Problem

Many customer-support interactions still depend on typing or navigating menus. This can create friction for users who prefer speaking, have difficulty typing, or want a more natural way to ask for help.

Proposed solution

VoiceMate AI is a prototype multilingual, voice-enabled customer-support assistant. A user can submit a spoken query or type a question. For the voice flow, Gnani AI Speech-to-Text converts the audio into text, Google Gemini generates a response, and Gnani AI Text-to-Speech turns the response into audio for playback. The application is presented through an interactive Streamlit dashboard.

How it works

1. **Input:** The user records a voice query or enters text.
2. **Transcription:** Gnani AI Speech-to-Text transcribes recorded audio.
3. **Response:** Google Gemini generates an answer from the transcribed or typed question.
4. **Voice output:** Gnani AI Text-to-Speech generates spoken audio for the answer.

Technology

Python for application logic; **Streamlit** for the user interface; **Gnani AI** for speech-to-text and text-to-speech; and **Google Gemini** for response generation.

Current status and scope

The Streamlit dashboard and the end-to-end English voice interaction have been tested by the developer. Language choices are included in the interface, but the accuracy and voice quality of every listed language have not yet been fully validated. VoiceMate AI is a prototype and is not presented as a production-ready customer-support deployment.

Next improvements

Validate each supported language with representative audio; improve error handling and response quality; add conversation history and configurable support knowledge; and evaluate latency, transcription quality, and user experience with structured tests.

Project

GitHub repository: github.com/harshithkumark01-byte/VoiceMate_AI
Developer: Harshith Kumar K.